



Anhedonia is Associated with Altered Striatal Neurophysiology and Function in Adolescents Varying in Levels of Depression

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BACKGROUND

- Anhedonia—the reduced capacity for pleasure—is a common¹ and debilitating² feature of adolescent depression.
- Existing antidepressant treatments—which primarily target serotonergic and glutamatergic systems—seldom mitigate symptoms of anhedonia, possibly due to its multifaceted nature (e.g., reward valuation, effort expenditure, motivational drive, hedonic satiation).
- Previous work has implicated altered reward (e.g., striatal) circuitry³, which undergoes protracted maturation through adolescence⁴, in anhedonia; however, striatal functioning and dopaminergic neurophysiology remain unclear.
- Measures derived from resting-state functional magnetic resonance imaging (rsfMRI) can non-invasively provide *in vivo* measurements of various aspects of striatal functioning and neurophysiology.

Hypotheses: (1) Lower striatal regional homogeneity (ReHo) (Fig. 1, path b) and lower dopamine function (T2*) intensity (Fig. 1, path c) will be associated with higher levels of anhedonia. (2) Striatal tissue iron will moderate this relationship, such that lower ReHo will be most strongly linked to high levels of anhedonia at low level of tissue iron.

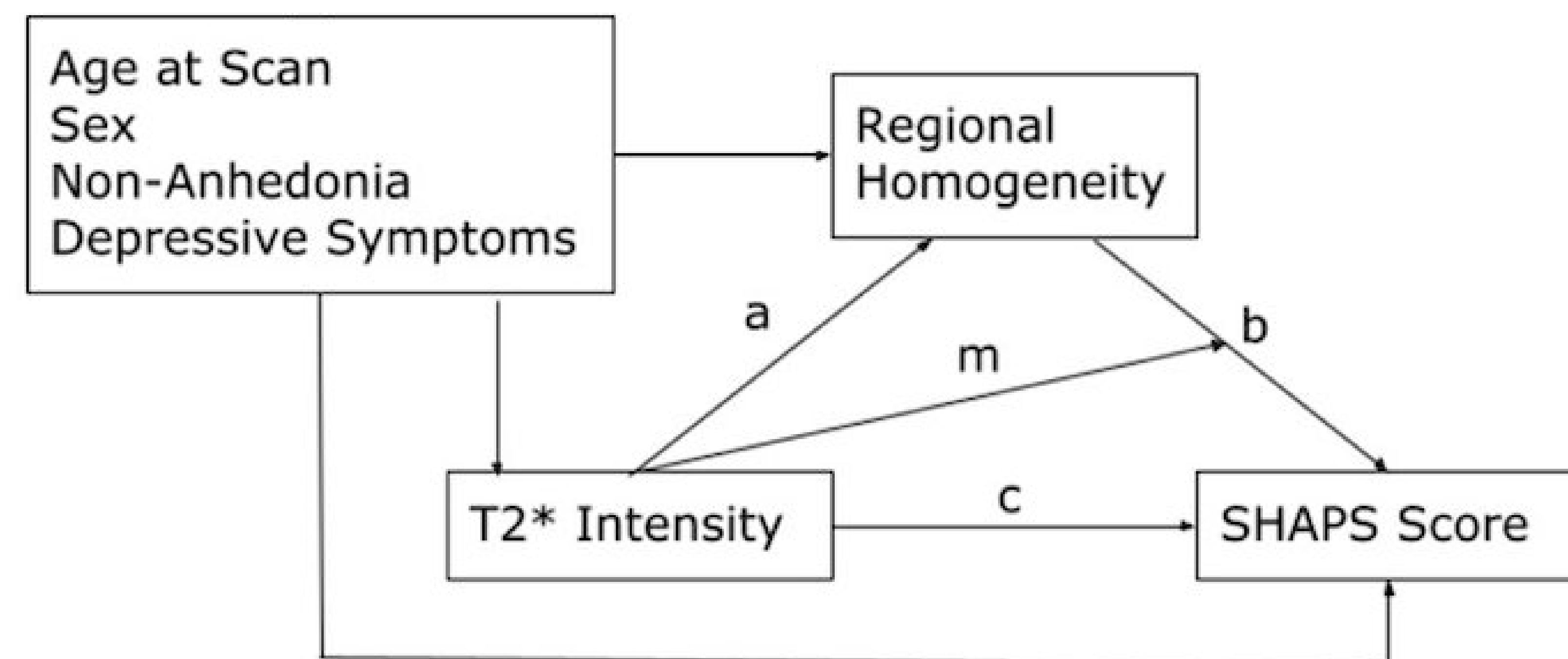


Figure 1. Path diagram for moderated mediation model.

METHODS

- Sample: 75 adolescents (M age = 15.30 (1.50), 46 F) participated in the study, of whom 56 scored ≥ 40 on the Children's Depression Rating Scale-Revised (CDRS-R)⁵ and 19 reported no current/past self/parent psychiatric diagnosis.
- Anhedonia was assessed using the Snaith-Hamilton Pleasure Scale (SHAPS)⁶ and other depressive symptoms using the Mood and Feelings Questionnaire (MFQ)⁷.
- We derived two measures from two 6-minute resting-state functional magnetic resonance imaging (rsfMRI) scans:
 - ReHo to assess regional voxel synchronization⁸
 - Normalized T2* signal to assess tissue iron⁹
- Data were preprocessed using *fmrprep*¹⁰.
- We used a voxel-wise moderated mediation approach to examine the relationships between T2* intensity, ReHo, and anhedonia severity, controlling for age, sex, and depressive symptoms besides anhedonia (Fig. 1).
- Clusterwise permutation testing accounted for multiple comparison corrections.

RESULTS

- Reduced striatal tissue iron in the left putamen was associated with higher levels of anhedonia (Fig. 2).
- Reduced ReHo was associated with higher levels of anhedonia in adolescents with higher striatal tissue iron in the right caudate, and with lower levels of anhedonia in adolescents with lower levels of tissue iron.

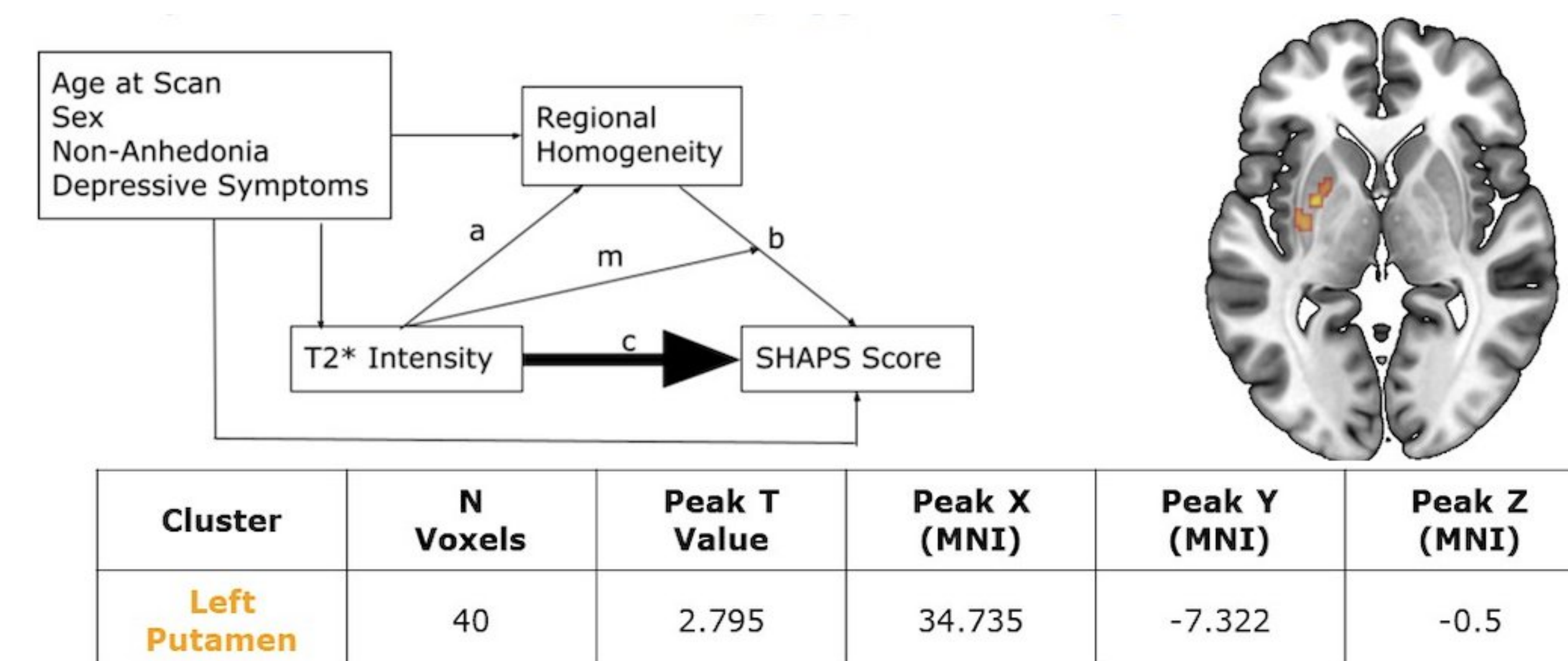


Figure 2. Simple slopes for T2* intensity effect on SHAPS score (Hypothesis 1)

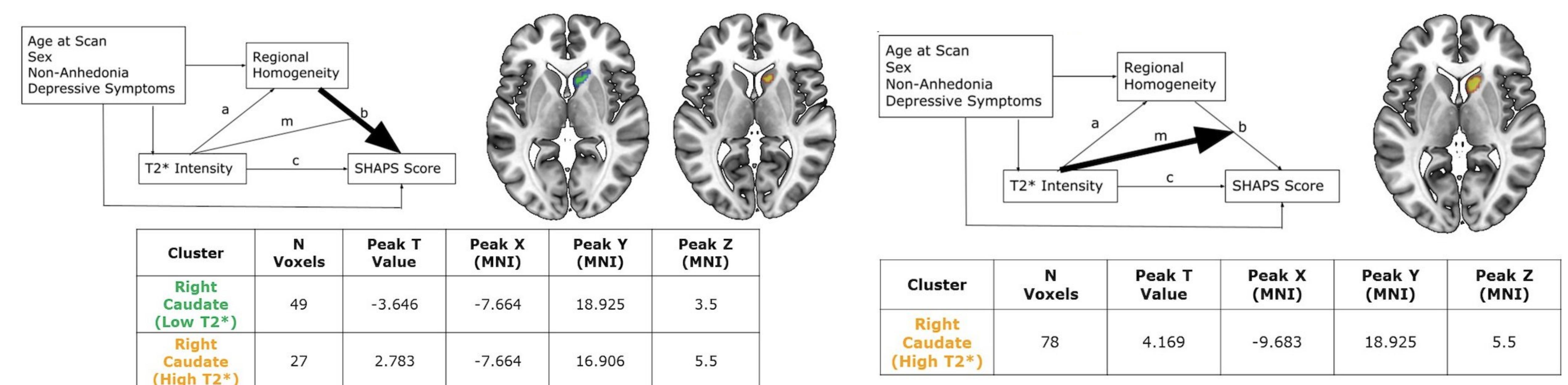


Figure 3. Simple slopes for ReHo effect on SHAPS score (Hypotheses 1 + 2)

Figure 4. Moderating effect of T2* intensity on ReHo-SHAPS score simple slopes (Hypothesis 2)

CONCLUSIONS

- Both lower ReHo and striatal tissue iron were associated with anhedonia beyond other depressive symptoms; these effects were specific to the dorsal striatum.
- Our findings suggest two distinct pathways to adolescent anhedonia: (1) higher dopamine availability in the putamen may contribute to motor-affective symptoms (e.g., motivational drive); (2) lower dopamine availability in the caudate along with disruptions in local synchronization may contribute to cognitive symptoms (e.g., updating reward contingencies).
- Future research is needed to determine the effectiveness of dopamine-targeted pharmacotherapy for adolescents with anhedonia and particularly those with altered dopaminergic functioning.

REFERENCES

1. Yorbik, O., Birmaher, B., Axelson, D., Williamson, D. E., & Ryan, N. D. (2004). Clinical characteristics of depressive symptoms in children and adolescents with major depressive disorder. *Journal of Clinical Psychiatry*, 65 (12), 1654-1659.
2. Ely, B. A., Nguyen, T. N., Tobe, R. H., Walker, A. M., & Gabbay, V. (2021). Multimodal investigations of reward circuitry and anhedonia in adolescent depression. *Frontiers in Psychiatry*, 12, 678709.
3. Forbes, E. E., Hariri, A. R., Martin, S. L., Silk, J. S., Moyses, D. L., Fisher, P. M., ... & Dahl, R. E. (2009). Altered striatal activation predicting real-world positive affect in adolescent major depressive disorder. *American Journal of Psychiatry*, 166 (1), 64-73.
4. Larsen, B., & Luna, B. (2015). In vivo evidence of neurophysiological maturation of the human adolescent striatum. *Developmental Cognitive Neuroscience*, 12, 74-85.
5. Poznanski, E. O., & Mokros, H. B. (1996). Children's depression rating scale, revised (CDRS-R)
6. Snaith, R. P., Hamilton, M., Morley, S., Humayan, A., Hargreaves, D., & Trigwell, P. (1995). A scale for the assessment of hedonic tone the Snaith-Hamilton Pleasure Scale. *The British Journal of Psychiatry*, 167 (1), 99-103.
7. Angold, A., & Costello, E. J. (1987). Mood and feelings questionnaire (MFQ). *Durham: Developmental Epidemiology Program, Duke University*.
8. Zang, Y., Jiang, T., Lu, Y., He, Y., & Tian, L. (2004). Regional homogeneity approach to fMRI data analysis. *NeuroImage*, 22(1), 394-400.
9. Haacke, E. M., Cheng, N. Y., House, M. J., Liu, Q., Neelavalli, J., Ogg, R. J., ... & Obenaus, A. (2005). Imaging iron stores in the brain using magnetic resonance imaging. *Magnetic Resonance Imaging*, 23(1), 1-25.
10. Esteban, O., Markiewicz, C. J., Blair, R. W., Moodie, C. A., Isik, A. I., Erramuzpe, A., ... & Gorgolewski, K. J. (2019). fMRIPrep: a robust preprocessing pipeline for functional MRI. *Nature Methods*, 16 (1), 111-116.

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